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MGCR 271 Final Project

What Drives Regular-Season Success in the NHL?

A Statistical Analysis of Regular Season Conference Winners (2015-2025, Excluding 2019-2020 and 2020-2021)

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Introduction

The National Hockey League (NHL) is one of the most competitive professional sports leagues. Topping the table in an 82-game season is a strategic challenge. As competition and spending increases across the Eastern and Western Conferences, determining what drives the regular season dominance has become increasingly difficult. While commentators often debate whether offense, defense, or goaltending matters most, our goal is to pursue a data-driven statistical approach that demonstrates which factors predict success. This report will study the correlation between PP%, PK%, (xGF - xGA), Penalty Drawn per Game, All Star Presence and coaching change of regular-season conference teams in the NHL from 2015 to 2025 with their total points earned, excluding the irregular 2019-2020 and 2020-2021 COVID-19 seasons to maintain data consistency. By studying eight seasons of team-level performance data in both conferences, we aim to uncover the correlation of these independent variables with the dependent variable, being the total points earned. The results will allow us to determine the significance of the independent variables to a successful team. By applying simple and multiple regression models supported by hypothesis testing, this study aims to identify the performance metrics that best predict NHL success. Through these statistical measures, we highlight patterns that separate top-performing teams from average ones and provide a framework that could help project future outcomes in a data-driven environment.

Statement of Hypotheses

Our estimation for the appropriate significance level (α) to evaluate our hypothesis is 5%. Because stronger performance in these variables is expected to contribute positively to winning games, all alternative hypotheses are right-tailed, predicting a positive effect on point totals.

$$H0: \beta 1 = 0 \text{ vs } Ha : \beta 1 > 0$$

Our null hypothesis for our All Stars variable states that there is no relationship total points earned in a Regular Season, while our alternative predicts that there is. Teams with All Star players tend to benefit from higher offensive production and team efficiency, increasing the likelihood of winning more games and accumulating more total points. Our null hypothesis for our Power Play Percentage (PP%) variable states that there is no relationship with total points, while our alternative

predicts that there is. A strong power play increases scoring efficiency during man-advantage situations, teams with higher PP% should convert more opportunities into goals and accumulate more total points. Our null hypothesis for our Expected Goals Differential (xGF – xGA) variable states that there is no relationship with total points, while our alternative predicts that there is. Expected goals differential captures the overall quality of its scoring chances and defense, where teams with high (xGF – xGA) control games and win more frequently, increasing total earned points. Our null hypothesis for our Penalty Kill Percentage (PK%) variable states that there is no relationship with total points, while our alternative predicts that there is. Teams with higher PK% allow fewer goals while shorthanded, improving defensive reliability in critical moments and increasing the likelihood of winning games and earning more points. Our null hypothesis for our Penalties Drawn Per Game variable states that there is no relationship with total points, while our alternative predicts that there is. Teams that consistently draw penalties tend to spend more time on the power play, which can translate into more goals and ultimately more wins. For our categorical variable, we examined whether experiencing a coaching change affects team performance. We hypothesize that instability behind the bench disrupts systems and consistency, which can negatively impact a team's regular-season results.

$$H0: \beta 1 = 0 \text{ vs } Ha: \beta 1 < 0$$

For the multiple regression model, we test whether our performance variables collectively explain differences in total points rather than acting independently. Our null hypothesis states that all coefficients equal zero, while the alternative proposes that at least one predictor has a significant effect on total points.

$$H0: \beta 1 = \beta 2 = \beta 3 = \beta 4 = \beta 5 = 0 \text{ vs } Ha: \text{at least one } \beta i \text{ is not } 0$$

Data Collection Process

To ensure that our analysis accurately reflects the factors that influence team success in the National Hockey League, we designed a structured and consistent data collection process. Our sample includes all NHL teams across eight regular seasons ranging from 2015–2016 to 2024–2025, while

deliberately excluding the 2019–2020 and 2020–2021 seasons due to the disruptions caused by the COVID-19 pandemic. These seasons were shortened and irregular, and therefore not comparable to standard 82-game schedules. Removing them improves consistency and reliability in both the descriptive and regression analyses. This results in 250 observations (31–32 teams per season, depending on league expansion).

The primary source for quantitative performance statistics was NHL.com, which provides official league records and ensures consistency with how statistics are reported across the league. From NHL.com's "Stats" and "Regular Season" sections, we retrieved the following continuous variables: total points earned (dependent variable), power play percentage (PP%), penalty kill percentage (PK%), expected goals differential (xGF – xGA), and penalties drawn per game. These statistics were collected for each team and each season by navigating to the corresponding season page and extracting the values directly from the league-wide standings and team statistical summaries. This also allowed us to ensure that all statistics corresponded to regular-season performance only. These were then sorted into an Excel spreadsheet, where a table was organized by sorting the variables into columns, with each team and their respective season into rows. Through this dataset, we were able to generate scatterplots for each independent variable, derive simple and multiple regression models using the Excel Data Analysis Toolpak.

Data Description

Our data contains eight Regular Seasons across a decade, totaling roughly 250 observations. This range provides enough variation to measure relationships between our independent variables and the dependent variable.

The dependent variable is total points earned in a regular season, which captures wins (2 points), overtime losses (1), and losses (0). Point totals range from the low 60s to over 120, with most teams between 85 and 105. The distribution is centered near the league average with a slight right tail caused by a small group of dominant teams; these are high values but not statistical outliers (See Appendix1). **PP% (Power Play %)** in the dataset ranges from about 12% to 33%. Most teams cluster in the

18–24% range, forming a clear unimodal distribution with a peak around 19.8–21.8%, where roughly 40–50 teams appear. A slight right tail reflects elite power-play units reaching the upper 20s and low 30s. Very few teams fall below 15%, indicating extremely weak power plays are uncommon. Teams in higher PP% intervals generally finish with more points, suggesting a positive association (See Appendix 1). **Penalties Drawn per Game** measures how often a team generates power-play opportunities. The variable is approximately normal and symmetric with a mean/median of 2.96 and a standard deviation of 0.25, ranging from 2.26 to 3.73. (See Appendix 1)When plotted against total points, the scatterplot shows a wide spread but a slightly upward-sloping trendline, indicating a weak positive relationship (See appendix 9). **Penalty Kill Percentage** is the independent variable and is fairly stable across seasons. PK% is tightly centered between 76% and 84%, forming a near-normal distribution. The histogram is symmetric with a single clear peak, reflecting league-wide consistency in penalty-killing performance. (See Appendix 1)The lack of long tails suggests that no teams had extreme shorthanded results. Because PK% is tightly clustered and goal totals have more spread, the data is suitable for testing how much PK% explains offensive output. Results show a moderately strong positive relationship: as PK% increases, goals scored also tend to increase, indicating teams with stronger shorthanded defense may generate more offense overall (See Appendix 7). **The variable Goals Differential (xGF – xGA)** ranges from approximately –25 to +25. The histogram has a smooth, bell-shaped curve centered near zero. (See Appendix 1)Teams with strong play-driving ability appear in the positive tail, while weaker teams fall into the negative tail. Although the distribution has slightly heavier tails than a perfect normal distribution, no values qualify as statistical outliers. Even without running regression, teams with higher differentials typically finish with more standings points, and those with negative differentials finish lower, showing xGF – xGA has a clear link to success(See Appendix 6) (See Appendix 1). **The Star Player variable** counts how many skaters on a team scored at least 85 points in a season. Values range from 0 to 4, but most teams have 0 or 1, making the distribution heavily right-skewed. Star players are rare but impactful: teams with one or more typically place higher in the standings, while teams without them tend to finish around average. This pattern supports the idea that elite scorers contribute significantly to team success(See Appendix 1). Finally, our **categorical variable identifies whether it was the coach's first full**

season with the team. About 37% of teams had a new coach during the seasons observed, highlighting frequent coaching turnover. While this alone may not strongly affect total points, we included the variable to test whether a categorical factor interacts with or influences the other variables (star players, penalties, goal differential, PK%, and PP%). (See Appendix 1)

Results and Analysis

In this section, we present and interpret the results for each independent variable, followed by the findings from the multiple regression model. Our goal is to evaluate how strongly each factor is associated with total points earned in a regular NHL season. For every independent variable, we analyze the linear regression trendline from the scatterplot, the coefficient of determination (R^2), the estimated slope coefficient, the 95% confidence interval for β_1, the t-statistic, and the p-value. We then determine whether to reject or fail to reject the corresponding null hypothesis.

A scatterplot of the number of star players (x-axis) against total points (y-axis), using 250 team-season observations, shows a clear upward trend: teams with more elite players tend to finish higher in the standings (Appendix 5). The fitted regression line, y = 8.6573x + 86.723, indicates a positive relationship, and the model's R^2= 0.1864 shows that about 18.6% of the variation in team points can be explained by a team's number of star players. The simple linear regression results support this association. The estimated slope coefficient, $\beta_1 = 8.66$, suggests that each additional star player is associated with an increase of roughly 9 standings points—a substantial effect in the context of competitive playoff races. The 95% confidence interval for β_1, [6.40, 10.92], lies entirely above zero, providing strong evidence that the effect is both positive and statistically significant. The t-statistic (t = 7.54) and the very small p-value ($p = 8.9 \times 10^{-13}$) allow us to reject the null hypothesis H_0: $\beta_1 = 0$ at the 5% significance level. This confirms that the number of star players significantly increases expected point totals. Overall, while star players do not explain all of the variation in team performance, the results demonstrate that teams with more elite talent reliably earn more points over the regular season.

A scatterplot of team PP% (x-axis) against total points (y-axis), using 250 team-season observations from 2015–16 to 2024–25 (excluding the two pandemic-affected seasons), displays a clear upward trend (Appendix 1). While point totals vary widely across teams, the fitted trendline shows a consistent positive relationship, indicating that teams with higher PP% generally finish with more standings points. The regression equation, y = 2.3977x + 42.494, and the corresponding R^2= 0.2936, suggesting that roughly 29% of the variation in team performance can be explained by a team's power-play efficiency. The simple linear regression output provides strong statistical evidence for this relationship. The estimated slope coefficient β_1 = 2.3977 indicates that a one-percentage-point increase in PP% is associated with an increase of approximately 2.4 standings points over an 82-game season, a meaningful effect given how slowly point totals accumulate. The 95% confidence interval, [1.93, 2.86], lies entirely above zero, confirming that the effect is both positive and statistically significant. The t-statistic (t = 10.15) is extremely large in magnitude, and the corresponding p-value (1.77×10^{-20}) is effectively zero. Therefore, at the 5% significance level, we reject the null hypothesis H_0: $\beta_1 = 0$. We conclude that PP% is a significant predictor of team success, and that higher power-play efficiency reliably corresponds to higher point totals.

A scatterplot of PK% and total points, based on 250 team-season observations, shows a clear upward trend (Appendix 7). Although point totals vary widely, the positive slope of the fitted trendline indicates that teams with stronger penalty kills generally earn more standings points. The regression equation, y = 2.1609x - 81.104, along with R^2 = 0.2111, shows that PK% explains about 21% of the variation in team performance, less than PP%, but still meaningful for a single statistic. The regression output confirms that this relationship is statistically significant. The estimated slope coefficient, β_1 = 2.1609, suggests that a one-percentage-point increase in PK% corresponds to roughly 2.16 additional standings points over a season. The 95% confidence interval for β_1, [1.64, 2.68], lies fully above zero, and the t-statistic (t = 8.15) is large in magnitude. The extremely small p-value (1.84×10^{-14}) is far below the 5% significance level. Therefore, we reject the null hypothesis H_0: $\beta_1 = 0$, concluding that PK% is a statistically significant predictor of team success. While PK% does not

capture all defensive factors, teams with stronger penalty-killing units consistently finish with higher regular-season point totals.

A scatterplot of expected goals differential (xGF-xGA) against total points shows a very clear upward trend across all 250 team-season observations from 2015-16 to 2024-25 (excluding the two COVID-affected seasons). The data points form a tight, positively sloped cluster, indicating that teams generating better scoring-chance quality than their opponents almost always finish higher in the standings (Appendix 6). This visual relationship is much stronger than what we observe for PP%, PK%, or penalties drawn. The fitted regression line is given by the equation y = 2.9104x - 54.762, with an R^2 of 0.7186, meaning that expected goals differential alone explains about 72% of the variation in team point totals, by far the strongest single predictor in our study. The estimated slope coefficient, $\beta_1 = 2.9104$, indicates that a one-unit increase in xGF - xGA is associated with roughly 2.9 additional standings points in an 82-game season. The effect size is substantial and consistent with what analytics research suggests: teams that create significantly more expected goals than they allow tend to dominate possession, generate higher-quality chances, and win more games. The 95% confidence interval for β_1, [2.23, 2.66], lies entirely above zero, providing strong evidence against the null hypothesis. The corresponding t-statistic (t = 25.17) is extremely large in magnitude, and the p-value (3.06×10^{-70}) is effectively zero. Therefore, at the 5% significance level, we reject H_0: $\beta_1 = 0$ and conclude that expected goals differential has a positive and statistically significant impact on total points. Overall, xGF - xGA is the single strongest performance indicator in our dataset. Its very high explanatory power and highly significant slope coefficient show that expected goals differential is closely linked to team success and is one of the clearest predictors of whether a team finishes near the top of the standings.

A scatterplot of penalties drawn per game (x-axis) against total points (y-axis), using 250 team-season observations from 2015-16 to 2024-25, shows a very mild upward trend (Appendix 9). The regression equation displayed on the chart, y = 13.707x + 50.397, and the corresponding R^2 = 0.0462, indicate that penalties drawn per game explain only about 4.6% of the variation in team point

totals. The cloud of points is widely dispersed, reflecting a weak overall association. The simple regression output supports this interpretation. The slope coefficient, $\beta_1 = 13.707$, suggests that increasing penalties drawn per game is associated with slightly higher point totals, although the practical effect size is small. (See A The 95% confidence interval for β_1, [5.92, 21.49], lies entirely above zero, indicating statistical significance. The t-statistic ($t = 3.47$) and the p-value ($p = 0.00062$) both show strong evidence against the null hypothesis. At the 5% significance level, we reject H_0: $\beta_1 = 0$ and conclude that penalties drawn per game have a statistically significant but weak positive relationship with total points. Compared with stronger predictors such as PP% or expected goal differential, penalties drawn contribute relatively little to explaining team success.

To evaluate whether a coaching change affects team performance, we ran a simple linear regression using a binary variable where 1 = coaching change and 0 = no coaching change. The scatterplot and fitted regression line indicate a slight downward shift for teams that changed coaches. The estimated regression equation is $y = -5.8094x + 93.19$ (Appendix 4). This means that teams without a coaching change are predicted to earn about 93.19 points, while teams with a coaching change average roughly 5.8 fewer points. This reflects the idea that coaching transitions, often occurring mid-season or after a poor previous year, tend to accompany weaker team performance. The slope coefficient, $\beta_1 = -5.81$, quantifies this effect and shows that coaching changes are associated with a modest decrease in expected point totals. The relationship is statistically significant: the t-statistic is $t = -2.789$, and the corresponding one-tailed p-value ($p \approx 0.00285$) is well below the 5% significance level. Because the 95% confidence interval for β_1 lies entirely below zero, we reject the null hypothesis H_0: $\beta_1 = 0$. However, the $R^2 = 0.0304$ indicates that coaching changes explain only about 3% of the variation in team points. While statistically significant, the effect is small in practical terms, and coaching stability is far from a major predictor of team success compared to performance metrics like PP%, PK%, or expected goals differential. Overall, the results suggest that teams undergoing coaching changes tend to earn fewer points, but the impact is limited, and most variation in team performance must be explained by other factors (Appendix 3).

Multiple Regression

Our statistical analysis examined how several team performance metrics contribute to NHL standings points, beginning with simple regressions before combining all predictors into a multiple regression model. After running the multiple regression with all six predictors (PP%, PK%, xGF%, Penalties Drawn, All Stars, and Coaching Change)(See Appendix 11), our fitted model becomes:

$$Points = -124.57 + 0.921(PP\%) + 0.836(PK\%) + 3.565(PenaltiesDrawn) + 1.071(All\text{-}Stars) + 2.397(\text{xGF} - \text{xGA}) - 2.313(Coaching\ Change)$$

The model achieved an R^2 of 0.8337, indicating that 83.37% of the variation in team points is explained by the six predictors. This is exceptionally high for real sports analytics data and suggests that these variables collectively provide a strong explanation of performance outcomes.

The Adjusted R^2 of 0.8296 confirms that even after accounting for the number of predictors and degrees of freedom, the model remains highly accurate and is not overfitted. This supports the reliability of the regression's predictive ability.

xGF% is by far the strongest predictor in the model. Teams that consistently generate higher-quality scoring chances outperform their opponents and accumulate significantly more points. PP% is also highly statistically significant. Effective special teams play directly contributes to scoring and momentum, confirming that power-play efficiency is a key component of success. Penalties Drawn per Game also remained significant in the multiple model ($\beta = 3.565$, $p = 0.043$), though its weaker effect size and borderline p-value suggest a smaller practical contribution, but we still reject the null hypothesis at the 5% level. Coaching Change had a significant negative effect ($\beta = -2.3128$, $p = 0.00999$), providing evidence to reject the null and indicating that teams switching coaches midseason tend to underperform, likely due to pre-existing issues and instability. Finally, the All-Stars variable was not statistically significant in the multiple regression ($\beta = 1.070$, $p = 0.088$), meaning we fail to reject the null hypothesis; having elite players did not independently improve point totals once team performance metrics such as xGF% and special teams efficiency were accounted for.

Our multiple regression indicates that the simple-regression effect of All-Star players was partially confounded by underlying team performance metrics such as expected goals differential and special teams efficiency. In isolation, All-Stars appeared strongly significant; however, once xGF – xGA, PP%, and PK% were included, its coefficient became statistically insignificant. This demonstrates that star players do not independently increase point totals—rather, their presence is correlated with strong systems, shot quality, and special teams play. This illustrates the importance of controlling for confounders to avoid overstating causal interpretations. Overall, the results show that NHL success is driven primarily by strong underlying play (xGF%), supported by effective power play and penalty kill units, while penalties drawn and coaching changes play smaller roles, and star presence alone does not significantly impact team outcomes.

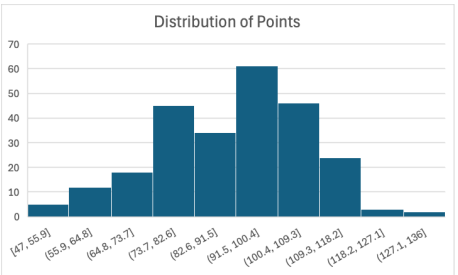
Recommendations

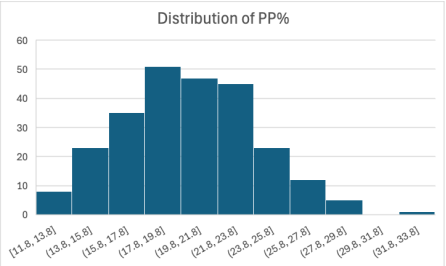
Based on our results, we recommend that teams place a strong emphasis on improving their expected goals differential, since it was the most powerful predictor of total points in our analysis. Teams that consistently created more good scoring chances and allowed fewer chances from their opponents finished much higher in the standings, showing that controlling play at five on five has the biggest impact on regular-season success. Special teams performance remains another actionable area. Because both PP% and PK% showed meaningful links to point production, even small improvements in power-play efficiency and penalty-kill structure can translate into tangible standings gains across an 82-game season. Teams should emphasize shot selection, zone entries, and movement on the power play, while also improving shorthanded pressure, clearing efficiency, and defensive support on the penalty kill. Marginal gains in these areas often decide tightly contested games. While penalties drawn were only weakly related to success, teams should still look for opportunities to attack aggressively and draw calls in dangerous areas. Creating more power-play opportunities is beneficial only if paired with an efficient PP, so drawing penalties should not be viewed as a standalone strategy, but rather as a way to amplify existing strengths. Finally, frequent coaching changes may be costly. Although the effect size was small, the negative association suggests that instability behind the bench can hinder

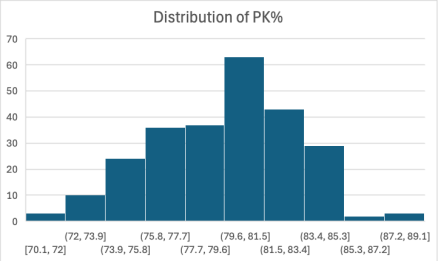
performance. Wherever possible, organizations should commit to long-term systems, rather than reacting to short stretches of poor results. Stability allows tactical structure to develop, and developing structure appears to drive point totals more than rapid change.

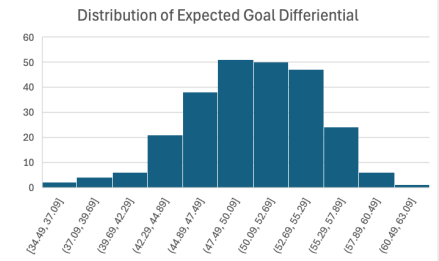
Appendices

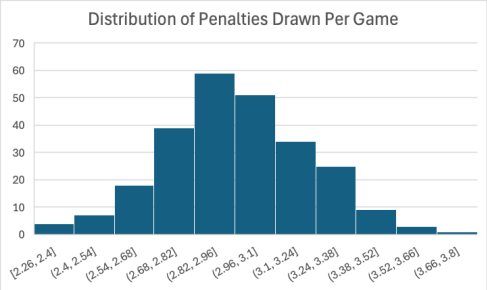
*Appendix 1:**Distributions of variables***

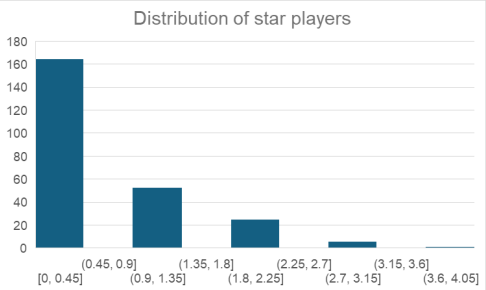


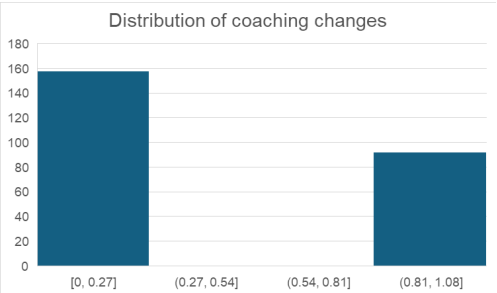




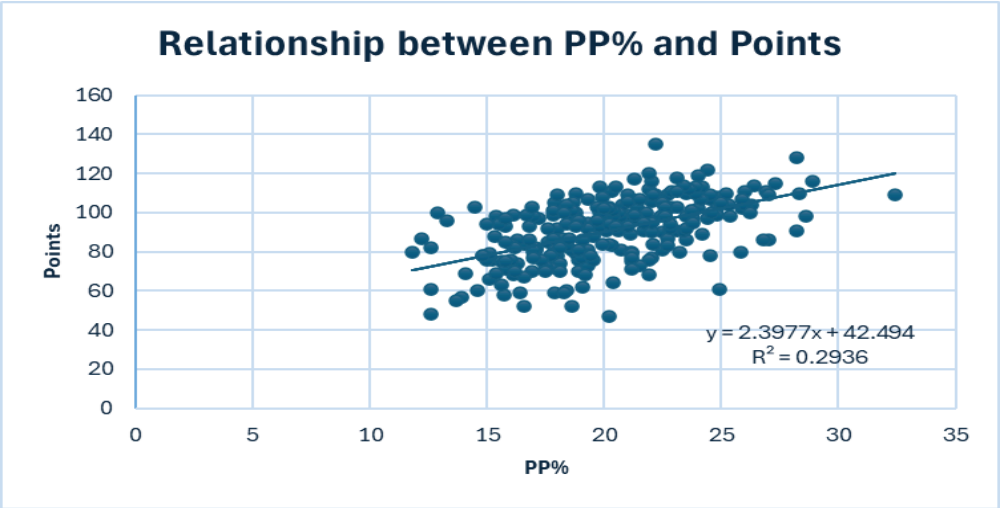








*Appendix 2: **PP%***



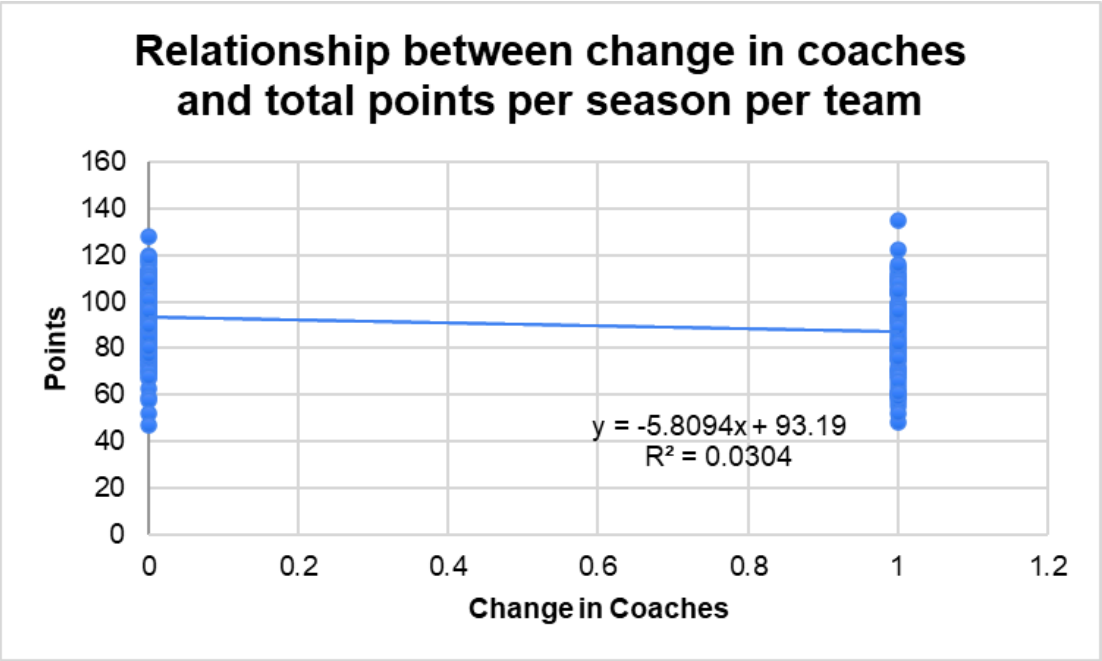
Appendix 3: ***Regression Summary for PP%***

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.54185842							
R Square	0.29361055							
Adjusted R Sc	0.2907622							
Standard Erre	13.5564049							
Observations	250							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	18943.8476	18943.8476	103.081119	1.7692E-20			
Residual	248	45576.4764	183.776115					
Total	249	64520.324						
	Coefficients	*Standard Erroı*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	42.4942869	4.85889412	8.74567048	3.4372E-16	32.9243273	52.0642464	32.9243273	52.0642464
X Variable 1	2.39767495	0.23615696	10.1528872	1.7692E-20	1.93254596	2.86280394	1.93254596	2.86280394

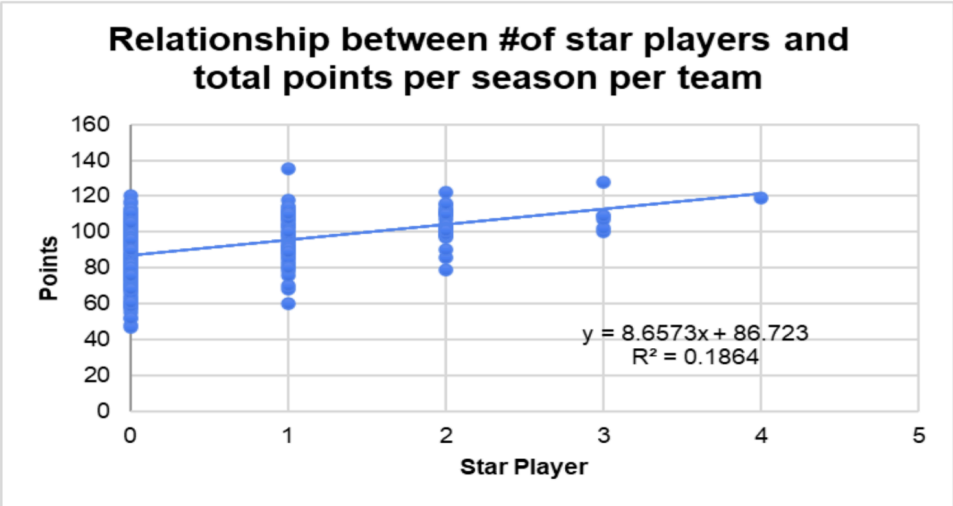
Appendix 4 ***(regression summary for coaching change):***

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.174396736							
R Square	0.030414221							
Adjusted R Square	0.026504601							
Standard Error	0.476781868							
Observations	250							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	1.768404	1.768404	7.779329	0.005694813			
Residual	248	56.3756	0.227321					
Total	249	58.144						
	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.844685591	0.173547	4.86718	2.02E-06	0.50287123	1.186499952	0.50287123	1.186499952
X Variable 1	-0.005235312	0.001877	-2.78914	0.005695	-0.008932266	-0.00153836	-0.008932266	-0.00153836

Appendix 4: ***Coach Change***



Appendix 5: ***Star Players***



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.431790182
R Square	0.186442761
Adjusted R	0.183162288
Standard E	14.54844198
Observatio	250

Relationship between #of all stars and points

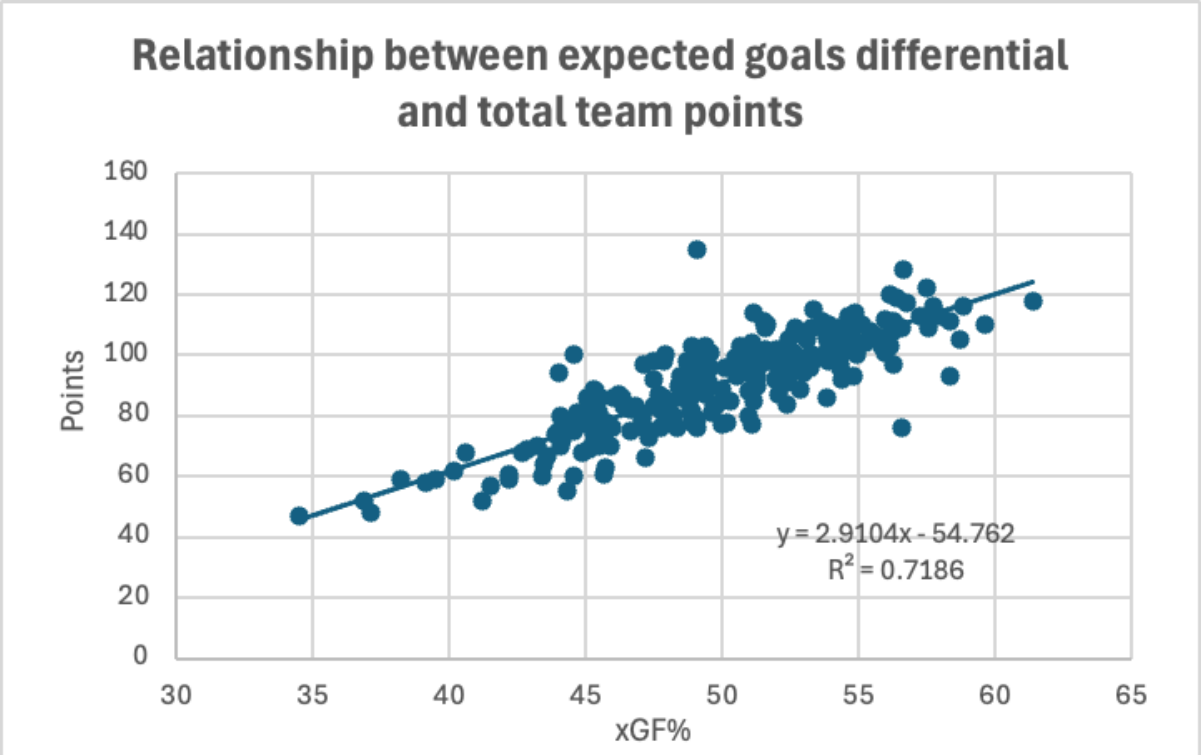
ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regressio	1	12029.34735	12029.34735	56.83411386	8.90948E-13
Residual	248	52490.97665	211.6571639		
Total	249	64520.324			

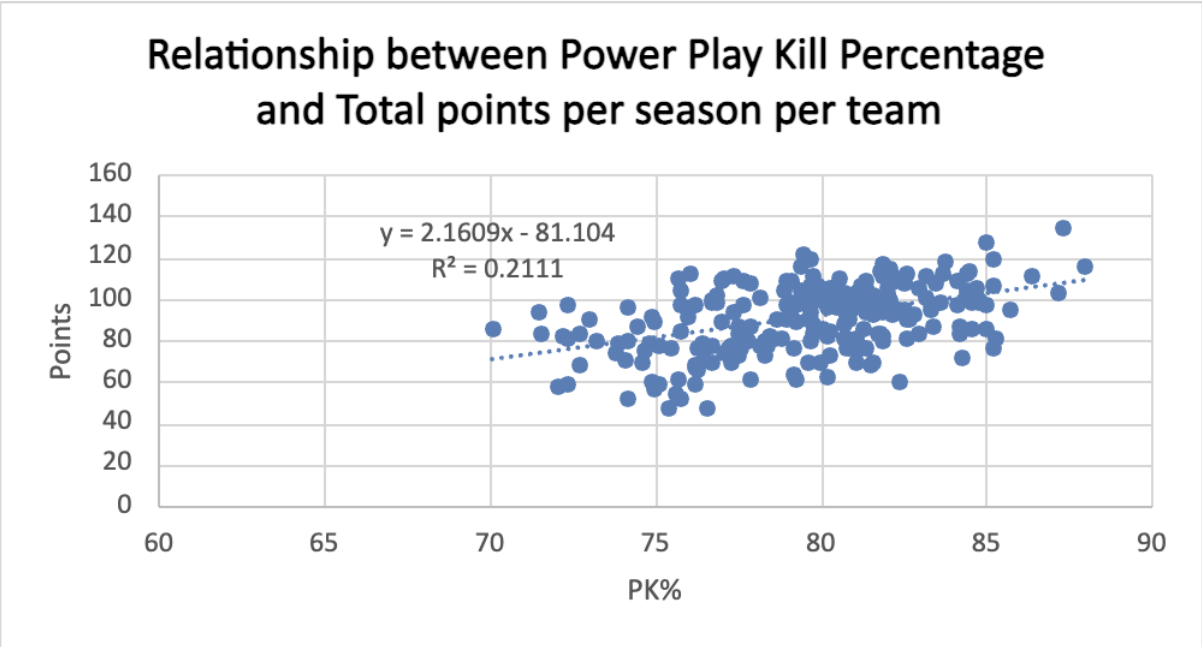
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	86.72333956	1.084579509	79.96033379	4.5943E-179	84.58717817	88.85950096	84.58717817	88.859501
X Variable	8.657320872	1.14836241	7.538840352	8.90948E-13	6.395534229	10.91910752	6.395534229	10.9191075

Appendix 6: ***(xGF - x GA), xGF%***

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.8477219							
R Square	0.71863241							
Adjusted R Sc	0.71749787							
Standard Erre	2.47399514							
Observations	250							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	3876.87768	3876.87768	633.409268	3.0572E-70			
Residual	248	1517.92169	6.12065196					
Total	249	5394.79936						
	Coefficients	*Standard Erroı*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	27.6118044	0.90625017	30.4681923	8.3657E-86	25.8268761	29.3967326	25.8268761	29.3967326
X Variable 1	0.24691853	0.00981096	25.1676234	3.0572E-70	0.2275951	0.26624196	0.2275951	0.26624196



Appendix 7: ***PK%***



Appendix 8: ***Summary output PK%***

SUMMARY OUTPUT

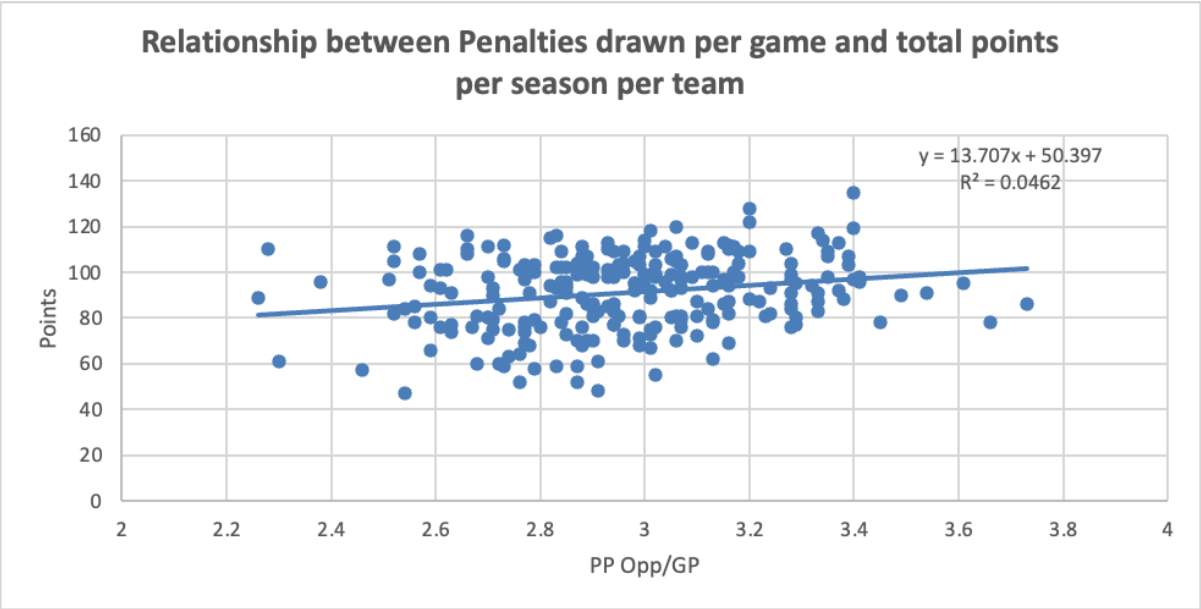
Regression Statistics	
Multiple R	0.459472148
R Square	0.211114655
Adjusted R Square	0.207933666
Standard Error	14.32614602
Observations	250

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	13621.18594	13621.18594	66.3676094	1.84278E-14
Residual	248	50899.13806	205.2384599		
Total	249	64520.324			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-81.10384833	21.1515658	-3.834413448	0.000159656	-122.7634574	-39.444239	-122.7634574	-39.4442393
X Variable 1	2.160861658	0.26524602	8.146631783	1.84278E-14	1.638439555	2.68328376	1.638439555	2.683283762

Appendix 9: ***Penalties Drawn Per Game***



Appendix 10: **Summary output Penalties Drawn Per Game**

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.215001
R Square	0.04622543
Adjusted R Square	0.04237957
Standard Error	15.752348
Observations	250

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	2982.47978	2982.47978	12.0195141	0.00062039
Residual	248	61537.8442	248.136469		
Total	249	64520.324			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	50.3965478	11.768931	4.28216867	2.6478E-05	27.2167478	73.5763479	27.2167478	73.5763479
X Variable 1	13.7073501	3.9537577	3.46691709	0.00062039	5.92012522	21.494575	5.92012522	21.494575

Appendix 10: ***Summary output Multiple Regression***

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.91307729
R Square	0.83371015
Adjusted R Sc	0.82960422
Standard Erro	6.64474025
Observations	250

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	53791.2488	8965.20813	203.050638	1.1413E-91
Residual	243	10729.0752	44.152573		
Total	249	64520.324			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-124.56613	10.8404747	-11.490837	1.079E-24	-145.91942	-103.21284	-145.91942	-103.21284
PP%	0.92050716	0.13859581	6.64166678	2.0144E-10	0.64750469	1.19350963	0.64750469	1.19350963
PK%	0.83631034	0.13514534	6.18822933	2.5658E-09	0.57010451	1.10251617	0.57010451	1.10251617
Penalties Dra	3.56518376	1.75339912	2.03329848	0.04310927	0.11138308	7.01898444	0.11138308	7.01898444
All Stars	1.07008328	0.62533844	1.71120662	0.08831903	-0.1616924	2.30185894	-0.1616924	2.30185894
xGF%	2.39711301	0.10362961	23.1315442	2.4082E-63	2.19298605	2.60123997	2.19298605	2.60123997
Coaching Ch	-2.3127702	0.89062056	-2.5968075	0.00998317	-4.0670917	-0.5584486	-4.0670917	-0.5584486

References and links:

- "NHL Stats, History, Scores, Standings, Playoffs, Schedule & Records." *Hockey*, www.hockey-reference.com/. Accessed 25 Nov. 2025.
- https://moneypuck.com/stats.htm